CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
SAN FRANCISCO BAY REGION

State Water Resources Control Board's
GENERAL WASTE DISCHARGE REQUIREMENTS
FOR SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS
ORDER WQ 2014-0153-DWQ

MONITORING AND REPORTING PROGRAM

for

Google Bay View Campus
Wastewater and Stormwater Treatment System

100 Lomax Lane
Mountain View, CA 94043

CIWQS Place No. 862046
GeoTracker Global ID WDR100052903
Waste Discharger ID (WDID) 2 43GWDR862046

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I. GENERAL

A. MONITORING AND REPORTING PROGRAM PURPOSE

This Monitoring and Reporting Program (Monitoring Program) describes requirements for monitoring the Google Bay View Campus Wastewater and Stormwater Treatment Systems (Facility) located at 100 Lomax Lane, Mountain View in Santa Clara County co-located at the NASA Ames Research Center. The Facility is owned and operated by Google LLC (Discharger). The Discharger is responsible for compliance with the monitoring and reporting requirements specified in this Monitoring Program. The Facility is subject to the Notice of Applicability (Notice) for regulatory coverage under the *General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems* Order WQ 2014-0153-DWQ (General Order), included as Notice Attachment C. This Monitoring Program applies to the wastewater and stormwater treatment systems and associated discharges to land via irrigation and recycled water supply for flushing fixtures (toilets and urinals), building and hardscape wash down, and cooling tower makeup within the central utility plant as described in the Title 22 Engineering Report and Report of Waste Discharge submitted for the Discharger's coverage under the General Order and authorized in the Notice.

Collected storm and rainwater will be routed through two onsite bioretention pond cells, an open water pond, and a retention pond followed by an onsite stormwater treatment system with a peak design flowrate of 50,000 gallons per day (gpd). The stormwater will be treated via an ultrafiltration membrane system with a membrane pore size of 0.02 micrometer (µm), ozonation, ultraviolet (UV) disinfection, and activated carbon.

Domestic wastewater will be collected and treated onsite to disinfected tertiary recycled water standards via a treatment system with a maximum design capacity of 85,000 gallons per day. Following a rotary drum 3-millimeter opening screen, the domestic wastewater will be pumped to two settling tanks totaling 26,000 gallons, equalized in tanks with a combined capacity of 60,000 gallons, and treated through a moving bed bioreactor process followed by polishing through a horizontal flow subsurface treatment wetland. The moving bed bioreactor aerobic biological treatment system consists of a pre-anoxic zone, blower aerated biochemical oxygen demand and nitrification zones, and clarification. Following the treatment wetland, the effluent is pumped back to the central utility plant for final treatment via a Schreiber's Fuzzy Filter compressible media filter, Aquionics In-line 400+ UV disinfection system unit with two reactors, granular activated carbon adsorption filters, and ozonation. The treated wastewater and stormwater are combined into a 60,000-gallon blend tank for storage prior to reuse.

The principle purposes of a Monitoring Program by the Discharger are:

1. To document compliance with the General Order, Notice, and Title 22 Engineering Report; and
2. To facilitate accountability and self-monitoring by the Discharger in the prevention and abatement of pollution or potential threats to water quality arising from waste discharges or recycled water use.

B. MONITORING AND REPORTING PROGRAM BASIS

Monitoring and reporting responsibilities of waste dischargers are specified in California Water Code (Water Code) sections 13267(b), 13268, and 13383. This Monitoring Program is issued pursuant to Water Code section 13267.

Water Code section 13267 states, in part:

> "… the regional board may require that any person who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge, waste outside of its region that could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports. In requiring those reports, the regional board shall provide the person with a written explanation with regard to the need for the reports and shall identify the evidence that supports requiring that person to provide the reports."

Water Code section 13268 states, in part:

> "(a) Any person failing or refusing to furnish technical or monitoring program reports as required by subdivision (b) of section 13267, or failing or refusing to furnish a statement of compliance as required by subdivision (b) of section 13399.2, or falsifying any information provided therein, is guilty of a misdemeanor and may be liable civilly in accordance with subdivision (b).
>
> (b)(1) Civil liability may be administratively imposed by a regional board in accordance with article 2.5 (commencing with section 13323) of chapter 5 for a violation of subdivision (a) in an amount which shall not exceed one thousand dollars ($1,000) for each day in which the violation occurs."

The monitoring reports are necessary to document the Discharger's compliance with the Notice, General Order, and the California Code of Regulations Title 22 (22 CCR) Water Recycling Criteria, and the burden imposed in their preparation bears a reasonable relationship to the need for the reports and the benefits to be obtained from them. More specifically, the monitoring data are needed to evaluate the performance of recycled water production, distribution, and use and to assure protection of public health and the environment. Inconsistently treated or misused recycled water could expose people to pathogens or other harmful constituents and could adversely affect beneficial uses. Pursuant to Water Code section 13267, the Discharger shall implement this Monitoring Program and submit the monitoring reports described herein.

The monitoring requirements and practices described in the Monitoring Program are, generally, minimum levels of monitoring. Additional monitoring, other than as specified here, may be necessary to assure proper performance of the recycled water treatment system, such as in response to treatment changes or investigations of uncertainties. If the Discharger monitors any pollutant at the locations designated herein more frequently than required by this Monitoring Program, using test procedures approved by the Code of Federal Regulations Title 40 (40 CFR), part 136, or as specified in this Monitoring Program, the results of such monitoring shall be included in the calculation and reporting of the data submitted in monitoring reports.

II. SAMPLING AND ANALYTICAL METHODS

1. All discharges to land and recycled water uses must be monitored by observations and measurements sufficient to adequately characterize the location, quantity, and quality of the discharges. Wastewater sources and treatment systems also must be monitored in

order to verify that adequate and reliable treatment and management of the wastewater is being provided.

2. The Discharger shall maintain a chain of custody form that records the name of the sampler, sample type (grab or composite), time, date, location, and any preservative used for each sample, available to the San Francisco Bay Regional Water Quality Control Board (Water Board) upon request. The chain of custody form must also contain all custody information including date, time, and to whom samples were relinquished.

3. Field test instruments may be used if they are used by a State Water Resources Control Board (State Water Board) California Environmental Laboratory Accreditation Program certified laboratory, or:
   a. The user is trained in proper use and maintenance of the instruments;
   b. The instruments are field calibrated prior to monitoring events at the frequency recommended by the manufacturer;
   c. Instruments are serviced and/or calibrated by the manufacturer at the recommended frequency; and
   d. Field calibration reports are maintained and available for at least three years.

4. Collecting composite samples is acceptable in most cases. However, due to short sample holding times, bacteriological samples collected to verify disinfection effectiveness must be grab samples.

5. Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity.

6. Flow measurement is the accurate measurement of the flow volume over a given period of time using a properly calibrated and maintained flow measuring device. Flow determination from a properly calibrated and maintained automated pump-use recording device, such as a pump dose event counter, for a calibrated and documented pump, is acceptable.

III. DEFINITION of TERMS

The following definitions and descriptions of terms apply to the requirements of this Monitoring Program.

1. **Sampling Frequency**

   Continuous = Continuous monitoring, for the full duration of the monitored event.
   Daily = One time each calendar day.
   Monthly = One time per calendar month, at intervals of about four weeks.
   Quarterly = One time per calendar quarter, at intervals of about three months.
   Annually = One time per calendar year.

   Continuous Verification Monitoring = Ongoing confirmation of system performance using sensors for continuous observation of selected parameters, including surrogate parameters that are correlated with pathogen $\log_{10}$ reduction target requirements.

2. **Description of Monitoring Stations**

   a. **Monitoring Station Definitions**. Stations used for sampling and observations in accordance with this Monitoring Program shall be described in each monitoring report submitted in response to this Monitoring Program. The station description shall include a description of the water, wastewater, point of the wastewater system, or land area where specified monitoring is conducted.

   b. **Monitoring Station Changes.** Changes to the monitoring stations defined in this Monitoring Program may be authorized by the Water Board Executive Officer in order to accommodate changes in the recycled water system or system operations or to provide improved monitoring. Requests for changes to the monitoring stations must be submitted to the Water Board in writing, with a detailed explanation of the purpose of the proposed station changes. Proposed changes to monitoring stations must be approved in writing from the Executive Officer prior to implementation.

   c. **Site Plan Showing All Monitoring Stations.** The Discharger shall develop a legible plan view drawing that clearly depicts the locations of all major components of the wastewater treatment system, all monitoring stations identified in this Monitoring Program, and relevant land use features such as buildings, access roads, property boundaries, and surface water drainage systems. A copy of this drawing shall be included with all reports submitted in response to this Monitoring Program.

IV. MONITORING REQUIREMENTS

A. SYSTEM OPERATION PROCESS CONTROL MONITORING

Title 22 Engineering Report section 2.8.1 identifies UV disinfection system parameters that will be continuously monitored for disinfection system process control and operational effectiveness even though the Monitoring Program does not require submittal of the data to the Water Board. For most parameters, continuous monitoring is recommended to demonstrate compliance with effluent water quality limitations, ultraviolet (UV) dose requirements, effectiveness of the treatment system, and that the recycled water is safe for the intended uses. The data collected can also be assessed to determine whether pathogen $\log_{10}$ reduction targets are being met (refer to section IV.I).

B. DOMESTIC WASTEWATER MONITORING

Influent Monitoring

1. In alignment with General Order, the nitrogen content of the influent wastewater and recycled water land applied via irrigation shall be monitored and assessed based on the parameter total nitrogen, since the discharge flow rate to irrigation may be greater than 20,000 gpd. If the flow rate to irrigation is continually less than 20,000 gpd, the Discharger can request a change in the influent nitrogen monitoring requirement.

2. Nitrogen Percent Reduction: Influent samples shall be collected from a location that provides representative samples of the domestic wastewater quality entering the domestic wastewater treatment system via the sewer mining lift station. The influent samples shall be collected at approximately the same time as effluent samples. Compare the influent total nitrogen value to the effluent concentration to determine the treatment system's total nitrogen percent reduction.

3. At a minimum, influent monitoring shall consist of the following:

Table 1: Influent Monitoring Requirements

Parameter	Units	Sample Type	Sampling Frequency	Reporting Frequency
Flow, wastewater influent to treatment system	gallons per day	Meter	Continuous	Annually [1]
Total Nitrogen [2]	mg/L-N	Grab	2 x / year Wet & Dry Season: • Collect influent & effluent total nitrogen samples concurrently • Collect 3 samples per sampling event	Quarterly

[1] The monthly influent wastewater flow values are reported annually via the State Water Board's GeoTracker Electronic Submittal of Information (ESI) system per section IV.C *Volumetric Reporting* of this Monitoring Program.

[2] The total nitrogen percent reduction shall be determined based on three influent total nitrogen samples collected concurrently during a wet season sampling event (generally from October to April) and three influent samples collected during the dry season (generally from May through September). The influent samples shall be collected concurrently with the effluent samples.

Effluent Monitoring

4. Since biochemical oxygen demand, total suspended solids, and total nitrogen reduction requirements are per the General Order for domestic wastewater, the samples shall be collected following treatment and prior to the blending tank. The UV disinfection system monitoring shall be conducted at the disinfection system.

5. Final combined treated recycled water samples shall be collected from a location following the final storage tank, representative of the final recycled water distributed to the uses.

6. If recycled water is discharged to the land surface or used for flushing fixtures (toilets and urinals), then the final effluent shall be sampled daily when recycled water is produced for distribution and analyzed for total coliform. The samples shall be taken after the UV disinfection system but prior to the granular activated carbon bed and chlorine injection point. This requirement does not apply to subsurface discharges of recycled water.

7. Monitoring of the recycled water shall consist of the requirements listed in Table 2.

8. Nitrate values shall be reported "as Nitrogen," in values of milligrams per liter of nitrogen.

Table 2: Recycled Water Monitoring Requirements

Parameter	Units	Sample Type	Sampling Frequency	Reporting Frequency
Flow, recycled water produced, prior to storage tank	**Refer to Notice section IV.C Volumetric Reporting**		Monthly	Annually
Flow, average daily discharged to land	gpd	Meter	Continuous	Quarterly
Flow, maximum daily discharged to land	gpd	Meter	Continuous	Quarterly
Flow, volume for indoor uses	gallons	Meter	Continuous	Quarterly: Report monthly volume
Total Dissolved Solids	mg/L	Grab	Monthly	Quarterly
Total Nitrogen	mg/L	Grab	Monthly	Quarterly
Nitrogen Forms (1)			Annually	Annually
Total Kjeldahl Nitrogen	mg/L	Grab / Calculate	Annually	Annually
Ammonia	mg/L	Grab	Annually	Annually
Nitrate-Nitrite	mg/L as N	Grab	Annually	Annually
Total inorganic nitrogen	mg/L	Calculate	Annually	Annually
Sample Treated Domestic Wastewater Prior to Blending Tank				
Biochemical Oxygen Demand$_5$	mg/L	Grab	Monthly	Quarterly
Total Suspended Solids	mg/L	Grab	Monthly	Quarterly
Total Coliform (2)	MPN / 100 mL	Grab	Daily during domestic recycled water production	Quarterly: • Report Daily Result • Report Exceedances of Limits in Table 3
Total Nitrogen % Reduction (3)	%	Calculate	2 x / year Wet & Dry Season: • Collect influent & effluent total nitrogen samples concurrently • Collect 3 samples per sampling event	Quarterly
Sample at the Domestic Wastewater Disinfection System				
Turbidity	NTU	Meter	Continuous	Quarterly: • Report Daily Maximum & 95th percentile • Report Exceedances of Table 3 Turbidity Limits

Parameter	Units	Sample Type	Sampling Frequency	Reporting Frequency
UV Transmittance	%	Meter	Continuous	Quarterly: • Report Daily Minimum • Report Exceedances of Limit in Table 3
UV Dose	mJ/cm^2	Meter / Calculate	Continuous	Quarterly: • Report Daily Minimum • Report Exceedances of Limit in Table 3
UV Intensity	mW/cm^2	Meter	Continuous	Reporting not required

1) Total Nitrogen = Total Kjeldahl Nitrogen (TKN) + nitrite (NO_2) + nitrate (NO_3). TKN is the sum of ammonia nitrogen (NH3) + organic nitrogen. Total inorganic nitrogen is the sum of nitrite + nitrate + ammonia nitrogen.
2) Changes to the total coliform sample collection frequency and UV disinfection system parameter reporting are dependent upon demonstration of treatment system performance and disinfection effectiveness per Monitoring Program sections I and V.A., Senate Bill 966 criteria, and Water Board Executive Officer approval.
3) The total nitrogen percent reduction values shall be calculated two times per year. The total nitrogen percent reduction values shall be calculated based on the three influent total nitrogen samples collected concurrently with the effluent samples during a wet season sampling event (generally from October to April) and three samples collected during the dry season (generally from May through September).

Table 3: Reporting Requirements for Disinfection System Exceedances

Parameter	Report Exceedance Value (1)
UV Transmittance	< 55%
UV Dose	< 100 mJ/cm^2 either reactor
Total Coliform	> 2.2 MPN / 100 mL rolling 7-day median > 23 MPN / 100 mL 30-day maximum if more than 1 exceedance occurs during 30-day period. Can exceed maximum limit once. > 240 MPN / 100 mL any time
Turbidity	> 2 NTU average within 24 hours > 5 NTU more than 5% in 24 hours > 10 NTU any time, maximum

1) Total coliform and turbidity limits are per 22 CCR sections 60301.230(b) and 60301.320(a)(2), respectively.

C. VOLUMETRIC REPORTING

1. In order to establish an estimate of statewide recycled water use and potential for increased recycled water use statewide, the Water Quality Control Policy for Recycled Water (Recycled Water Policy) requires Dischargers to report the volume of treated

recycled water.[1] Annual reporting is necessary to evaluate progress towards the recycled water goals in the Recycled Water Policy and evaluate the need to update the recycled water goals in the future based on consistent statewide data. Additional information on the recycled water volumetric reporting is accessible online.[2]

2. The Discharger shall monitor monthly recycled water volumes treated onsite and **report annually** to the State Water Board's online **GeoTracker database by April 30** of each calendar year. The data shall be submitted to the GeoTracker database under the site-specific global identification number **WDR100052903**. Refer to Notice section V.D.4 *Electronic Reporting Format* for additional information on submitting via GeoTracker.

3. The Discharger shall report annually in accordance with each of the applicable items in section 3.2 of the Recycled Water Policy as described below:

   a. Influent.
   Monthly volume of wastewater collected and treated by the wastewater treatment plant.

   b. Production.
   Monthly volume of wastewater treated, specifying level of treatment.

   c. Discharge.
   Monthly volume of treated wastewater discharged to land, where beneficial use is **not** taking place, specifying level of treatment.

   d. Reuse.
   Monthly volume of recycled water distributed.

   e. Reuse Categories.
   Annual volume of treated wastewater distributed for beneficial use in compliance with 22 CCR in each of the applicable use categories listed below:

      1) Landscape irrigation

      2) Commercial application: commercial facilities, business use (such as laundries and office buildings), car washes, retail nurseries, and appurtenant landscaping that is not separately metered.

      3) Industrial application: manufacturing facilities, cooling towers, process water, and appurtenant landscaping that is not separately metered.

      4) Other non-potable uses: including but not limited to dual plumbed flushing fixtures, dust control, flushing sewers, fire protection, fill stations, snow making, and recreational impoundments.

   Note that the Recycled Water Policy includes other reuse categories that are not applicable to the Facility and are not stated herein.

[1] The Water Quality Control Policy for Recycled Water is accessible online at: https://www.waterboards.ca.gov/board_decisions/adopted_orders/resolutions/2018/121118_7_final_amendment_oal.pdf
[2] Volumetric reporting website: https://www.waterboards.ca.gov/water_issues/programs/recycled_water/volumetric_annual_reporting.html

D. SUBSURFACE WASTEWATER TREATMENT CELLS MONITORING

1. The subsurface wastewater treatment cells shall be visually inspected **weekly**, at a minimum, for proper operation including the following:[3]
   a. Maintenance of flow uniformity (inlet and outlet structures), which include indicators that the flow is backing up, ponding, or causing flow cavities.
   b. Pumps and flow controls. All pump controls and flow distribution devices and controls shall be inspected for proper operation as recommended by the manufacturer.
   c. Vegetation health
   d. Odor control
2. A **summary of inspections** that resulted in corrective actions shall be submitted with the next regularly scheduled monitoring report.

E. FINAL BLEND TANK MONITORING

1. All components and appurtenances of the final blend tank, including but not limited to, risers, valves, and air gaps, shall be inspected **weekly**, at a minimum, for proper operation by a qualified service provider, vendor, or Facility operator.
2. A **summary of inspections** that resulted in corrective actions shall be submitted with the next regularly scheduled monitoring report.

F. LAND APPLICATION AREA MONITORING

1. The Discharger shall monitor land application areas **monthly, at a minimum**, when recycled water is applied. If recycled water is not applied during a reporting period, then state so in the monitoring report. Land application area monitoring shall include the following:

Table 4: Land Application Area Monitoring Requirements

Parameter	Sample Type	Monitoring Frequency	Reporting Frequency
Observation of land application areas Refer to Monitoring Program section IV.F.2	Observation	Monthly, minimum	Quarterly

2. **Land Application Area Observations**

   The following observations shall be conducted **monthly, at a minimum,** when recycled water is being discharged to land. Refer to the Standard Observations template at the end of this Monitoring Program.

   a. Check discharge system control equipment (e.g., valves, pumps, meters, etc.) for proper functioning. Note operational status during observations (e.g., on/off; in use/not in use).

[3] Source: Environmental Protection Agency. September 2000. Constructed Wetlands Treatment of Municipal Wastewaters, Chapter 6. EPA/625/R-99/010.

b. Check all irrigation emitters for proper operational status (e.g., proper height, direction, and trajectory).

c. Check for evidence of recycled water escaping the discharge areas and offsite through uncontrolled surface runoff or through airborne spray. If recycled water escape is observed, show the source and affected area on a site plan drawing.

d. Check perimeter drainage ways and other runoff control systems for proper condition.

e. Check for odor from discharge areas. If nuisance odors are observed, describe the odor, indicate apparent source or cause, direction of travel, and any public use area or offsite area affected by the odors.

f. Check for evidence of recycled water ponding, and/or evidence of mosquitoes breeding within the discharge area due to ponded water. Note the course of action taken in response to recycled water ponding.

g. Check the soil for impacts related to the accumulation of salts, which can be indicated by the following symptoms: White crust on soil surface, water-stressed plants, leaf tip burn, poor drainage, black powdery residue on soil surface, and grey-colored soil.

h. Check that warning signs are properly posted to inform public that irrigation water is recycled water that is not safe for drinking.

G. SOLIDS MONITORING

1. The Discharger shall report **annually** the handling and disposal of all solids (e.g., screenings, sludge, biosolids, etc.) generated at the wastewater and stormwater treatment systems.
2. All grease interceptors shall be inspected, and serviced if needed, as frequently as necessary to assure continued proper operation.
3. Primary settling tanks shall be inspected **weekly** per the Operations and Maintenance Plan and pumped when it is determined necessary.

H. DUAL PLUMBED RECYCLED WATER SYSTEMS

The frequency of testing for cross-connection and backflow prevention devices shall be as listed in Table 5.

Table 5: Dual Plumbed Recycled Water Systems

Requirement	Frequency	Reporting Frequency
Cross-Connection Testing	Four Years [(a)]	30 days / Annually [(b)]
Backflow Incident	--	24 hours from discovery
Backflow Prevention Device Testing and Maintenance	Annually [(c)]	Annually

a) Testing shall be performed at least every four years, or more frequently at the discretion of the Division of Drinking Water.
b) Cross-connection testing shall be reported pursuant to 22 CCR, section 60314. The report shall be submitted to DDW within 30 days and included in the monitoring report to the Water Board.
c) Backflow prevention device maintenance shall be tested by a qualified person as described in 17 CCR, section 7605.

I. PATHOGEN REMOVAL / INACTIVATION INDICATORS & MONITORING

The Water Environment & Reuse Foundation (WE&RF) publication entitled *Risk-based Framework for the Development of Public Health Guidance for Decentralized Non-potable Water Systems* provides wastewater treatment system risk-based fecal pathogen $\log_{10}$ reduction targets for non-potable uses of waters onsite. The use of risk-based pathogen reduction credits is an attempt to achieve a specific level of human health protection, typically expressed as a tolerable burden of disease.[4]

Log_{10} reduction is defined by the National Blue Ribbon Commission (NBRC)[5] as the removal of a pathogen or surrogate in a unit process expressed in $\log_{10}$ units. A 1-log reduction equates to 90% removal, 2-log reduction to 99% removal, 3-log reduction to 99.9% removal, etc. Log_{10} reduction targets were established for specified pathogen groups (e.g., viruses, bacteria, or protozoa) to achieve an agreed level of risk to individuals of an annual "tolerable" infection risk of 10^{-4} infection per year.[4] Meeting these $\log_{10}$ reduction targets is dependent upon proper wastewater treatment system design, construction, operation, and maintenance. Pathogenic microorganism control is achieved by meeting $\log_{10}$ reduction targets, which are verified through measurements of influent and effluent quality. In addition, the Facility includes continuous monitoring systems for detecting a surrogate parameter that correlates directly with $\log_{10}$ reduction values for one or more of the reference pathogen groups.

Pathogen crediting frameworks generally consist of requirements for validation, field verification, and ongoing monitoring of treatment performance. For continuous $\log_{10}$ reduction value verification of process performance, WE&RF stated that it is common to use non-biological surrogates monitored with instrumentation. The NBRC-published *A Guidebook for Developing and Implementing Regulations for Onsite Non-potable Water Systems* that states the recommended standard for pathogen removal and/or inactivation is to continuously monitor using microbial, chemical, or physical indicator(s) or surrogate parameter(s) that verify the performance of each treatment process's ability to achieve its credited pathogen removal and/or inactivation. Table 4 of the Guidebook includes example continuous treatment process monitoring methods.[6]

Senate Bill 966 requires the State Water Board's establishment of water quality criteria for onsite treated non-potable water systems by December 1, 2022. Following the adoption of the Senate Bill 966 onsite reuse criteria, if the Discharger decides to continuously monitor surrogate parameters and the results provide assurance and demonstrate that the wastewater treatment system is achieving the log reduction target values, the total coliform sampling frequency may be modified upon the Water Board's Executive Officer's approval.

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[4] Source: Water Environment & Reuse Foundation. 2017. Risk-Based Framework for the Development of Public Health Guidance for Decentralized Non-Potable Water Systems. Chapter 3. Risk-Based Pathogen Reduction Targets.

[5] The National Blue Ribbon Commission for Onsite Non-potable Water Systems is comprised of 30 representatives from municipalities, public health agencies including the State Water Resources Control Board, water utilities, and national organizations. It is convened by the US Water Alliance, the Water Environment & Reuse Foundation, and the Water Research Foundation.

[6] Other National Blue Ribbon Commission for Onsite Non-potable Water System resources are available online at http://www.uswateralliance.org/initiatives/commission/resources.

If the Discharger decides to implement pathogen log_{10} reduction surrogate monitoring, then the Discharger shall submit to the Water Board a data-informed report that addresses the following:

1. Expected pathogen log_{10} reduction targets and credits for the treatment system processes.
2. Sampling plan acceptable to the Water Board Executive Officer to address how the treatment system's surrogate monitoring parameters and sample locations will be used to collect data to verify the pathogen log reduction credits.
3. Once the data are collected, compile the data to provide verification of the treatment system's effectiveness in meeting the pathogen log reduction requirement based on operating data across treatment stages.

V. REPORTING REQUIREMENTS

A. MODIFICATION OF MONITORING PRACTICES

1. Modifications of the monitoring and reporting practices specified in this Monitoring Program may be authorized by the Water Board Executive Officer, in consideration of acceptable accumulated data and alternate means of monitoring. Factors to be considered include the following: data quality, adequate characterization of the identified water or wastewater system process, consistency of system performance, compliance with waste discharge requirements, and acceptable means for providing equivalent and adequate monitoring of the identified water or wastewater system process.

2. Requests for modification of monitoring frequencies must be submitted to the Water Board in writing, with a technical report that includes evaluation of accumulated data, and a complete description of proposed alternate means of monitoring.

3. Proposed modifications of monitoring practices must be approved in writing from the Executive Officer prior to implementation.

B. INCREASED MONITORING FREQUENCY

1. If any monitoring indicates a violation of waste discharge requirements or unstable wastewater system operation or performance, or if any specified samplings or analyses are not completed as required, then the monitoring for the parameter(s) and monitoring station(s) in concern shall immediately and henceforth be conducted at twice the frequency identified in **Table 2** of this Monitoring Program. This increased monitoring frequency shall be maintained for at least two sampling events and until such time as the results of monitoring indicate violations are no longer occurring or the problem has been corrected and the wastewater system has returned to stable operation and performance.

2. The Discharger shall discontinue the distribution of recycled water to the indoor uses during any period in which it has a reason to believe that the quality of the recycled water is not meeting the Title 22 Uniform Statewide Recycling Criteria specifications.

C. REPORTS OF VIOLATIONS

In the event the Discharger violates or threatens to violate the conditions of the waste discharge requirements, the Discharger or Discharger's agent(s) **shall notify the Water Board office by email and telephone** as soon as the Discharger or Discharger's agent

has knowledge of the incident. Written notification shall be submitted **within 10 business days** of the date of the incident, unless directed otherwise by Water Board staff.

The written notification shall include pertinent information explaining reasons for the non-compliance and shall indicate what steps were taken to correct the problem and the dates thereof, and what steps are being taken to prevent the problem from recurring.

Contact Information

San Francisco Bay Regional Water Quality Control Board
Contact: Melissa Gunter or Margaret Monahan
Email: Melissa.Gunter@waterboards.ca.gov
Margaret.Monahan@waterboards.ca.gov
WDR.Monitoring@waterboards.ca.gov
Phone: (510) 622-2390 or (510) 622-2377
Mail: 1515 Clay Street, Suite 1400, Oakland, CA 94612

D. MONITORING REPORTING

The Discharger shall submit monitoring reports to the Water Board that document the Facility's operation and performance, and compliance with the General Order, Notice, and Title 22 Engineering Report in accordance with the following:

1. **Reporting Frequency and Schedule**
   Monitoring reports shall by submitted quarterly for each calendar quarter to the Water Board by the **first day of the second month following the three-month monitoring period**, as stated in Table 6.

Table 6: **Monitoring Data Reporting Schedule**

Due Date	Report	Reporting Period
May 1	1st Quarter	January 1 through March 31
August 1	2nd Quarter	April 1 through June 30
November 1	3rd Quarter	July 1 through September 30
February 1	4th Quarter	October 1 through December 31

2. **"As-Built" Plans of Completed Wastewater System**

   The Discharger shall submit complete plan-view design drawings of the completed wastewater system and all its components to the Water Board. The plans shall be submitted no later than **30 calendar days after finalization of the as-built plans** if changes made during construction are not included in the plans previously submitted to the Water Board.

3. **Treatment System Commencement Notification**

   The Discharger shall notify the Water Board upon commencement of the operation of the treatment system resulting in distribution to the recycled water uses. This notification shall be submitted no later than **5 calendar days** after the treatment system is operational.

4. **Electronic Reporting Format**

   a. When reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., influent, effluent, etc.), and reported analytical results for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with the General Order and indicate temporal trends, as applicable.

   b. The Discharger shall submit all reports and analytical data to the online State Water Board GeoTracker database via the site-specific GeoTracker Global ID **WDR100052903**.

   c. In order to submit reports electronically, create a secure online GeoTracker Electronic Submittal of Information (ESI) account, which will be connected to the Global ID. The Discharger can request a username and password online, which is accessible from the 'Getting Started' section on the GeoTracker ESI webpage.

      Log into the GeoTracker ESI system to upload reports and enter volumetric data.[7]

   d. Additional GeoTracker support information can be found at the following:

      1) 'Guides/Resources' document link in the "Tools" on the GeoTracker ESI account

      2) Resources on the GeoTracker ESI website, such as the ESI Beginner's Guide for Responsible Parties

      3) General GeoTracker Help Desk contact information:

         Phone: 1-866-480-1028
         Email: geotracker@waterboards.ca.gov

      4) Contact for reporting-related questions other than GeoTracker: WDR.Monitoring@waterboards.ca.gov

   e. GeoTracker volumetric annual reporting guidance documents are accessible online:

      1) Help Guide for Volumetric Annual Report in GeoTracker.

[7] GeoTracker password request: https://geotracker.waterboards.ca.gov/esi/password1.asp
GeoTracker ESI webpage:
https://www.waterboards.ca.gov/water_issues/programs/ust/electronic_submittal/
GeoTracker login: https://geotracker.waterboards.ca.gov/esi/login.asp
Beginner's Guide for Responsible Parties:
https://www.waterboards.ca.gov/ust/electronic_submittal/docs/geotracker_esi_rp_beginners_guide_revisedoct2019.pdf
Volumetric Reporting Help Guide:
https://www.waterboards.ca.gov/ust/electronic_submittal/docs/var_helpguide.pdf
Volumetric Reporting FAQ:
https://www.waterboards.ca.gov/water_issues/programs/recycled_water/docs/2020/faq_var.pdf

2) Frequently Asked Questions: Volumetric Annual Reporting of Wastewater and Recycled Water

5. **Monitoring Report Content**

Each monitoring report shall include, at a minimum, the following components:

a. Transmittal letter

A transmittal letter shall accompany each monitoring report submitted. The letter shall include the following:

1) Identification of:

a) Facility name and address;

b) Monitoring period being reported;

c) Violations of the General Order, Notice, or Title 22 Engineering Report requirements, and

d) Name and telephone number of a person familiar with the report and treatment systems, for follow-up discussions.

2) The transmittal letter shall contain the following penalty of perjury statement and shall be signed by the Discharger or the Discharger's authorized agent:

"I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment."

b. Summary table of the influent and effluent monitoring results, and observations, such as, but not limited to, the following:

1) Date of sampling or observation,

2) Location of sampling or observation (sample station),

3) Parameter of analysis,

4) Result of the analysis, measurement, or observation **from the current reporting period and previous reporting periods** for the influent and effluent monitoring,

5) Laboratory reports indicating the recycled water quality analytical results, and

6) Summary of wastewater treatment cells or final blend tank inspections that resulted in corrective actions.

c. A comparison of monitoring data to the discharge specifications and applicable effluent limits.

d. Facility site plan diagram that depicts the Facility, locations of all key wastewater treatment system components, and stations where sampling or observations occur.

e. Discussion of any violations of the General Order, Notice, 22 CCR, or the Title 22 Engineering Report, that addresses the following:

1) Date and time of occurrence;

2) Location of occurrence indicated on a plan drawing of the Facility and/or site and, if applicable, affected land areas;
3) Description of the violation; and
4) Corrective actions taken or planned to correct the problem, including but not limited to, increased monitoring, treatment system equipment or operation modifications. If a report describing corrective actions and/or a time schedule for implementation of actions was previously submitted, reference to the report is satisfactory.

VI. MONITORING PROGRAM CERTIFICATION

I, Eileen White, Executive Officer, hereby certify that this Monitoring and Reporting Program is effective on the following date: July 27, 2022.

for

Eileen White
Executive Officer

LAND APPLICATION AREA STANDARD OBSERVATIONS TEMPLATE

General Waste Discharge Requirements For
Small Domestic Wastewater Treatment Systems
Order WQ R2-2014-0153-DWQ Monitoring and Reporting Program

This template is optional but is intended to aid the Discharger in reporting the standard land application area observations required in Monitoring Program **section IV.F**. The tables may be reconstructed, and figures and additional information may be added.

The following observations shall be conducted **monthly,** at a minimum, when recycled water is discharged to land. Indicate if the answers are not applicable by entering 'N/A'.

Land Application Area Observations	
1. Check discharge system control equipment (e.g., valves, pumps, meters, etc.) for proper functioning. Note operational status during observations (e.g., on/off; in use/not in use).	☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____
2. Check all irrigation emitters for proper operational status (e.g., proper height, direction, and trajectory).	☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____
3. Check for evidence of recycled water escaping the discharge area and offsite through uncontrolled surface runoff or through airborne spray. If recycled water escape is observed, show the source and affected area on a site plan drawing.	☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____
4. Check perimeter drainage ways or other runoff control systems for proper condition.	☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____
5. Check for odor from discharge areas. If nuisance odors are observed, describe the odor, indicate apparent source or cause, direction of travel, and any public use area or offsite area affected by the odors.	☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____
6. Check for evidence of recycled water ponding, and/or evidence of mosquitoes breeding within the discharge areas due to ponded water. Note the course of action taken in response to recycled water ponding.	☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____

Land Application Area Observations
7. Check the soil for impacts related to the accumulation of salts. Circle the symptoms that apply. a. White crust on soil surface. Water-stressed plants. Leaf tip burn. b. Poor drainage. Black powdery residue on soil surface. c. Grey-colored soil. ☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____
8. Check that warning signs are properly posted to inform public that irrigation water is recycled water which is not safe for drinking. ☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____ ☐ Inspection Date _____